Linear systems and elimination

Near-quadratic solution cost at a prescribed backward error

Difficulty: challenging **Importance:** broadly interesting

Status: Solved

Rating rationale (historical): Challenging reflects the remaining removal of a dimension-dependent logarithm from an established general algorithm; broad impact follows from a condition-independent cost bound for arbitrary linear systems.

Resolution: affirmative, 12 September 2026

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Theorem 1, proved in Sections 2–6 gives a randomized exact-real algorithm with deterministic worst-case cost $O(n^2\varepsilon^{-3})$ and $O(n^2)$ scalar storage. For every original input it always returns a nonzero vector and achieves the original A-only backward error at most $5\varepsilon/8$ with probability greater than 0.997. Thus it settles the complete target with $q = 3$, uniformly in dimension, conditioning and right-hand side. Proof source.

The argument passed a separate independent Codex AI-agent audit, including exact component-test reproduction. This is informal automated review, not external human peer review or formal verification. The supplied draft was prepared with ChatGPT assistance; no Lean verification was performed.

Entrywise randomized rounding and weighted-pattern matrix products cancel the logarithmic iteration overhead in total arithmetic cost. This does not assert dimension-independent black-box convergence, finite-precision stability, bit complexity or practical speed. The original target and prior-source attribution below are retained; the earlier status discussion is historical.

Problem statement

Does there exist a randomized algorithm and absolute constants $C, q > 0$ such that, for every $n \geq 1$, nonsingular $A \in \mathbb{R}^{n\times n}$ with $\|A\|_2 = 1$, $b \neq 0$, and $0 < \varepsilon < 1/2$, it returns $x \neq 0$ satisfying

$$\Pr\left\{\frac{\|Ax - b\|_2}{\|A\|_2\|x\|_2} \leq \varepsilon\right\} \geq 0.99$$

using at most $Cn^2\varepsilon^{-q}$ operations? Use an exact-real arithmetic model with scalar arithmetic, square roots, comparisons, and independent standard Gaussian or random-bit draws at unit cost. All input entries may be accessed; the norm normalization is an input promise. The cost constants must be independent of the condition number and of n, b, ε. The fraction is normwise backward error with perturbations allowed in A only. A deterministic algorithm meeting the bound would also answer positively.

References and status

M. Dereziński, Y. Nakatsukasa, and E. Rebrova, *Towards Universal Convergence of Backward Error in Linear System Solvers* (22 May 2026), §§1, 3.1, 5.2, 7, poses the cost question and specifies the backward-error metric. Corollary 17 gives $O(n^2/\sqrt{\varepsilon})$ for PSD systems. The revised paper's Corollary 25 proves the general-system bound $O(n^2 \log(n/\delta)/\varepsilon)$ with failure probability δ; its §7 asks for convergence independent of dimension. The remaining target here removes the $\log n$ overhead, which cannot be hidden in a constant depending only on ε. The model and success probability above make this explicit. Searches on 2026-09-08 for "universal backward error 2026 linear systems", "MINBERR smoothed analysis", and "quadratic backward error linear solver" found no resolution of the displayed bound. The earlier version's lack of a proved general-system rate is no longer current.

Audit update — 2026-09-10

Rechecked version 2, Corollaries 17 and 25 and §7. Positive definite inputs satisfy the displayed cost by Corollary 17; the general bound still contains $\log n$. This warrants partial status and a difficulty change from extreme to challenging for the narrowed remaining gap. Searches for universal backward-error convergence and MINBERR follow-ups found no dimension-independent general bound.